Type MS · Sequence and describe the stages of mitosis

Answer 1 of 6 · Question 5 part b · 5 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

number				
5	(a) (i)	A (DNA); B/C/ and D are found in or associated with other structures.		1
	(ii)	correct circles of pair;		1
	(iii)	only two pairs of homologous chromosomes/ would be 23 in a human cell;		1
	(b)	• prophase, metaphase, anaphase and telophase; • (prophase) chromosomes become thicker/visible; • (metaphase) chromatids line up at equator; • (anaphase) chromatids separate and migrate to poles; • (telophase) nucleus/nuclear membrane reforms;	Allow condense Allow chromosomes throughout Allow clear annotated diagrams showing stages	5

Total 8 marks

Type MS · Sequence and describe the stages of mitosis

Answer 1 of 6 · Question 5 part b · 5 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Sequence and describe the stages of mitosis

Answer 2 of 6 · Question 6 part a.iv · 3 marks

2020 NOV · Paper 01 (regional R) · find part a.iv in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Sequence and describe the stages of mitosis

Answer 3 of 6 · Question 7 part a.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
7 (a)	(i)	• prophase, metaphase, anaphase and telophase;;	minus 1 for each error	2 marks
	(ii)	• chromosomes lined up; • equator shown/aligned at equator; • one member of each bivalent above, the other below;		3 marks
(b)	(i)	A (DNA); B not part of DNA C not part of DNA D found only in RNA not DNA		1 mark
	(ii)	length = 20 mm; $\frac{20}{750}$ = 0.026 mm; = 26.0/26.7 µm;	allow 20 -22 ECF	3 marks
(c)		• two genetically identical cells; • diploid/2n;		2 marks

Total 11 marks

Type MS · Sequence and describe the stages of mitosis

Answer 3 of 6 · Question 7 part a.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Sequence and describe the stages of mitosis

Answer 4 of 6 · Question 7 part a.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
7 (a)	(i)	• prophase, metaphase, anaphase and telophase;;	minus 1 for each error	2 marks
	(ii)	• chromosomes lined up; • equator shown/aligned at equator; • one member of each bivalent above, the other below;		3 marks
(b)	(i)	A (DNA); B not part of DNA C not part of DNA D found only in RNA not DNA		1 mark
	(ii)	length = 20 mm; $\frac{20}{750}$ = 0.026 mm; = 26.0/26.7 µm;	allow 20 -22 ECF	3 marks
(c)		• two genetically identical cells; • diploid/2n;		2 marks

Total 11 marks

Type MS · Sequence and describe the stages of mitosis

Answer 4 of 6 · Question 7 part a.ii · 3 marks

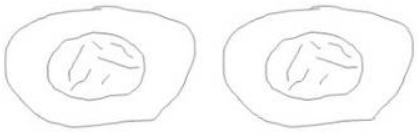
2021 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Sequence and describe the stages of mitosis

Answer 5 of 6 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a) (i)	156 – 72; 84;	ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark	2
(ii)	**C** 9 minutes;		1
(iii)	In the following order only: • oxygen; • diffusion; • aerobic/cellular; • decreases/reduces/goes down;		4
(b) (i)	 must have 6 chromosomes shape not important	One mark for each correct cell drawn	2
(ii)	**B** metaphase; A is incorrect as chromosomes do not align at the equator during anaphase C is incorrect as chromosomes do not align at the equator during prophase D is incorrect as chromosomes do not align at the equator during telophase		1
			Total 10

Type MS · Sequence and describe the stages of mitosis

Answer 5 of 6 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

uestion number	Answer	Notes	Mark s

Type MS · Sequence and describe the stages of mitosis

Answer 6 of 6 · Question 2 part b.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

2 (a)		**C (endoplasmic reticulum)** The answer is not A as the cell membrane transports substance into and out of the cell B cytoplasm as this is a base for chemical reactions D as ribosomes provide sites for protein synthesis		1
(b)	(i)	Z Y W X;		1
	(ii)	growth/repair;	Ignore repair of body cells	1
(c)		Any two from • mitosis produces diploid cells/46 chromosomes, meiosis produces haploid cells/23 chromosomes; • mitosis produces 2 (daughter) cells, meiosis produces 4 cells/gametes; • (daughter) cells from mitosis are genetically identical, (daughter) cells from meiosis show variation;		2
			Total for Question 2 = 5 marks	